12.	A	At the top: net force = centripetal force $\frac{mv^2}{r} = mg + N$ Normal force N > 0 for ball to remain inside pail.

		$\frac{mv^2}{r} > mg$ $v > \sqrt{rg} = \sqrt{0.7 \times 9.81} = 2.6$ m s $^{-1}$
13	D	Earth's surface $g = GM/R^2$